ZCAS UNIVERSITY

MASTER OF SCIENCE IN INFORMATION TECHNOLOGY
MASTER OF SCIENCE IN COMPUTER SCIENCE

CAS 6181

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

FINAL EXAMINATION

MONDAY 25th NOVEMBER 2024

TIME: 14.00 – 17.00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer Two (2) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

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SECTION A: QUESTION 1 IS COMPULSORY AND MUST BE ATTEMPTED

Question 1:

Scenario: Read the scenario in Appendix I (also sent to you earlier), and then answer the following questions:

a) Describe how Genetic Algorithms can be used to optimize the path planning of an autonomous drone. (10 Marks)

b) Explain the primary components of a Genetic Algorithm used in this case study?

(5 Marks)

c) Assume the fitness function for a flight path f(x) is given by:

$$f(x) = \frac{1}{1 + E(x)}$$

where $E(x)$ represents the total energy consumption for path x. If path x_1 has $E(x_1) = 50$ units and path x_2 has $E(x_2) = 30$ units, compare their fitness values.

(5 Marks)

d) Explain how Bayesian inference can help the drone manage uncertainty when navigating through a storm. (5 Marks)

e) Differentiate between Propositional Logic and First Order Logic in the context of drone AI. (5 Marks)

f) Construct a logical statement in First Order Logic that the drone could use to identify safe landing zones. (5 Marks)

g) Describe a fuzzy inference system that the drone could use to evaluate landing zones.

(10 Marks)

h) Explain how Fuzzy Logic improve the drone's decision-making compared to binary logic? (5 Marks)

[Total: 50 Marks]

SECTION B: ATTEMPT ANY TWO (2) QUESTIONS FROM THIS SECTION

Question 2:

Refer to the Mini Case Study below and then answer the questions that follow:

Mini Case Study: "Using Swarm Intelligence to Optimize Traffic Flow in Smart Cities"

Overview

Traffic congestion is a major problem in urban environments, causing delays, increased pollution, and energy waste. To address this issue, smart cities are exploring Swarm Intelligence principles to manage and optimize traffic flow. Inspired by the behavior of social insects like ants and bees, Swarm Intelligence leverages decentralized, self-organizing algorithms that enable traffic management systems to make real-time adjustments. By deploying a network of smart traffic lights and autonomous vehicles that communicate and respond like a swarm, cities aim to reduce congestion, improve travel times, and enhance road safety.

Case Study Details

In a pilot project, a city uses Swarm Intelligence algorithms to control a network of smart traffic lights. Each traffic light acts as an autonomous agent that makes decisions based on local conditions, such as vehicle density and traffic flow. The traffic lights communicate with each other to coordinate and minimize congestion. Similarly, autonomous vehicles equipped with sensors and communication modules adapt their routes based on real-time traffic information shared across the swarm.

The system is designed to balance traffic across the city efficiently, ensuring that heavily congested areas receive priority while maintaining smooth traffic in less busy zones.

a) Describe the main principles of Swarm Intelligence, and how they apply to traffic management in smart cities? (10 Marks)

b) Explain how the concept of "stigmergy" applies to Swarm Intelligence in traffic systems? (5 Marks)

c) Describe how the waggle dance concept from Bee Intelligence can be used to optimize drone communication in agriculture. (5 Marks)

d) What metrics would you use to evaluate the effectiveness of a Bee Intelligence algorithm in traffic system? (5 Marks)

[Total: 25 Marks]

Question 3:

Refer to the Mini Case Study below and then answer the questions that follow:

Mini Case Study: HealthNet and NeuroDiag

HealthNet is a healthcare technology company that has developed an advanced diagnostic system called NeuroDiag, which utilizes artificial neural networks (ANNs) to assist in diagnosing diseases from medical images. NeuroDiag processes various types of medical imaging data, including MRI and CT scans, to identify patterns and anomalies indicative of conditions such as tumors, fractures, and other pathologies. The system is trained on a large dataset of labeled medical images, enabling it to learn and recognize complex features associated with different diseases. HealthNet employs a multi-layered feedforward neural network with backpropagation for training the model. The network architecture includes an input layer, several hidden layers, and an output layer that provides the probability of different diagnoses.

NeuroDiag also incorporates a mechanism to handle missing data and employs dropout during training to prevent overfitting. The system has shown promising results in preliminary trials, with accuracy levels comparable to expert radiologists. However, continuous evaluation and retraining are required to maintain performance as new data becomes available. HealthNet is working on improving the interpretability of the neural network's decisions to gain the trust of medical professionals and ensure regulatory compliance.

a) Explain the structure of a feedforward neural network and how it applies to NeuroDiag. Describe the role of each layer in the network. (10 Marks)

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b) Describe the backpropagation algorithm used in NeuroDiag. How does it adjust the weights in the neural network? (10 Marks)

c) Given a simple neural network with one hidden layer of three neurons, input layer of two neurons, and an output layer of one neuron, calculate the total number of weights and biases in the network. (5 Marks)

[Total: 25 Marks]

Question 4:

Refer to the Mini Case Study below and then answer the questions that follow:

Mini Case Study – Robotics and the Role of Generative AI

Examples of robotics are everywhere nowadays, ranging from self-driving cars, industrial robots, civilian and military drones, human robots with human-like intelligence (such as ASIMO), and many more others. In terms of military robots, we note the destruction that drones have caused in Ukraine because of the on-going Russia-Ukraine war. The originators of such robots may not have intended for them to be used for the said destruction, but the advances of artificial intelligence have driven us to these levels.

A major area of advancement in artificial intelligence in Generative AI, with tools being used in various areas, essay writing, image creation, programming among others. Thus, programming of robots such as drones have become much easier. Most developed countries have developed military robots that are expected to be used for combat in future. Fear is however mounting among most of the world as "Killer robots" without so much of the human reasoning brain could be unleashed on the innocent populace everywhere. It is hoped that we will never reach that far.

a) Explain Isaac asimov's laws in relation to robots. (10 marks)

b) Discuss the major components of a robot and explain their relationship to intelligent agents. (10 Marks)

c) Briefly explain the common types of robots and indicate their application areas. (5 Marks)

[Total: 25 Marks]

APPENDIX 1

Case Study: "Optimizing Autonomous Drone Navigation Using Artificial Intelligence Techniques"

Abstract

Autonomous drone technology has made significant strides in recent years, largely due to advancements in artificial intelligence (AI) and machine learning (ML). This case study examines how AI techniques—specifically Evolutionary Computation/Genetic Algorithms, Uncertainty Management, Propositional/First Order Logic, and Fuzzy Logic—can be utilized to optimize autonomous drone navigation in uncertain environments. The goal is to create a robust and adaptable navigation system capable of making intelligent decisions even when data is incomplete or ambiguous.

Background

Drones are increasingly used for various applications, including surveillance, delivery, agriculture, and search-and-rescue operations. However, autonomous drones must handle complex challenges like dynamic environments, obstacles, and uncertain conditions such as fluctuating wind patterns or signal loss. An effective navigation system requires sophisticated AI algorithms that can optimize paths, manage uncertainty, make logical inferences, and handle ambiguous data.

Section 1: Evolutionary Computation / Genetic Algorithms

Description and Application:

Evolutionary Computation, specifically Genetic Algorithms (GAs), are inspired by the process of natural selection. These algorithms are used to solve optimization problems by iteratively improving candidate solutions based on a fitness function. In the context of drone navigation, GAs can optimize flight paths by finding routes that minimize energy consumption, avoid obstacles, and reduce travel time.

Problem Scenario:

Imagine a delivery drone needs to navigate a city to deliver packages efficiently. The drone's navigation system uses Genetic Algorithms to evolve flight paths, considering constraints like battery life, delivery time windows, and no-fly zones.

Section 2: Uncertainty Management

Description and Application:

Uncertainty Management involves techniques to deal with incomplete, imprecise, or noisy data. Probabilistic models, such as Bayesian networks, are commonly used to make informed decisions under uncertainty. In our case study, the drone faces uncertainties like unpredictable weather conditions, GPS inaccuracies, and changing obstacles.

Problem Scenario:

Suppose the drone must decide whether to continue on its path or reroute due to sudden wind gusts detected by its sensors. Using Bayesian inference, the drone calculates the probabilities of safe passage versus rerouting based on weather data and sensor readings.

Section 3: Propositional / First Order Logic

Description and Application:

Logical reasoning helps the drone make decisions based on predefined rules and observed facts. Propositional Logic deals with true or false values, while First Order Logic adds quantifiers and relationships among objects. The drone's AI uses logical reasoning to interpret sensor data and make rule-based decisions, such as landing when battery levels are critically low.

Problem Scenario:

For example, the drone's AI is programmed to recognize the rule: "If the battery level is below 20% AND no charging station is within 1 km, THEN initiate landing protocol." Using logical statements, the AI continuously evaluates conditions to ensure safe operation.

Section 4: Fuzzy Logic

Description and Application:

Fuzzy Logic allows the drone to handle ambiguous or imprecise information. Unlike traditional binary logic, fuzzy logic assigns degrees of truth between 0 and 1. For instance, when assessing whether a landing area is "safe," the drone might consider factors like terrain smoothness and proximity to obstacles using fuzzy sets.

Problem Scenario:

The drone assesses potential landing sites using criteria such as "smoothness" and "obstacle density." These parameters are not binary; instead, they fall on a spectrum. The AI system uses fuzzy rules to combine these inputs and decide on the best landing site.

END OF MAIN CASE STUDY